

ADNAN SAEED

AI Engineer

saeedadnan151@gmail.com | +92 315 5682476 | [linkedin.com/in/adnan-saeed-ai](https://www.linkedin.com/in/adnan-saeed-ai) | github.com/AdnanSaeed-85

SUMMARY

AI Engineer with 1+ year of professional experience designing and shipping production-grade agentic AI systems. Specialized in LangGraph, LangChain, RAG/CRAG, and multi-agent orchestration, with hands-on work in LLM tool calling, state-graph routing, prompt engineering, and memory management (in-context, vector store, and persistent via PostgreSQL). Built full-stack AI applications using FastAPI, Next.js, and PostgreSQL, deployed across Vercel, Cloudflare, and Hugging Face Spaces. Comfortable with Docker, Git, and cloud environments.

EXPERIENCE

AI Engineer — DZone Technologies

March 2026 – Present

- Designed and shipped LLM-powered agentic workflows using LangGraph and LangChain, including state-graph routing, tool calling, and multi-agent coordination
- Managed token budgets via chunking and summarization strategies for long-context meeting transcription (83MB, 29-min, 8-speaker meeting processed in under 2m 20s)
- Engineered system prompts, structured output specs, and few-shot examples to improve summarization reliability across Groq and Gemini models
- Implemented long-term memory via pgvector and short-term in-context state management for agentic session flows
- Integrated LLMs with external REST APIs (AssemblyAI, DeepL, YouTube Data API, Resend) via function/tool calling patterns
- Cut infrastructure costs by choosing AssemblyAI, Groq, and Gemini as cheaper alternatives and eliminating cloud storage via client-side audio extraction
- Deployed and maintained production services across Vercel, FastAPI Cloud, and Neon (PostgreSQL); handled cross-platform deployment challenges and CORS/async fixes

Stack: LangGraph, LangChain, FastAPI, Groq, Gemini, OpenAI, PostgreSQL, pgvector, Docker

AI Engineer — Developers360

July 2025 – November 2025

- Contributed to a Marketing Automation Agent: scraped Google Maps for target companies, extracted contact pages, and auto-filled outreach forms using Playwright and GPT-4o-mini via LangChain
- Built and tested multi-step LLM pipelines with structured output validation and error-recovery layers

PROJECTS

Advanced Blog Writing Planner Agent

Python, LangGraph, LangChain, RAG, LongTermMemory, ShortTermMemory, Tavily Search, Streamlit

- Built as a multi-agent blog writing system that takes a topic and produces a fully structured publication-ready blog
User enters topic -> AI plans structure and defines sections -> dedicated agent assigned to each section -> all agents work in parallel -> final agent merges all sections into a single Markdown file
- Integrated Tavily Search for real-time web research per section with automatic deduplication
- Built on Streamlit for topic input and blog management

BriefRoom — AI Meeting Assistant

Next.js, FastAPI, PostgreSQL, AssemblyAI, Groq, Gemini, LangChain, LangGraph

- Built and shipped as a solo full-stack web app that eliminates manual meeting note-taking
Upload video file or paste YouTube link -> transcription starts (queued -> processing with speaker diarization) -> AI generates summary -> if AI Prompt enabled, actionable prompt generated alongside -> final results ready to view
- Processed an 83MB, 29-minute meeting with 8 speakers in under **2 minutes 20 seconds end-to-end**

- Supports 4 user roles Guest, Signed-up, Premium, and Admin with JWT-based auth and role-based access
- Handled production challenges: storage cost migration, LLM token limit chunking, and cross-platform deployment
- Deployed on Vercel (frontend), FastAPI Cloud (backend), and Neon (PostgreSQL)

Live Demo: briefroom.app

Reddit Lead Scout — AI-Powered Lead Generation System

React, FastAPI, PostgreSQL, PRAW, OpenAI, WebSockets

- Built and shipped as a full-stack automated lead generation system for sales and business teams
User defines target subreddit + keywords -> system monitors Reddit continuously -> OpenAI filters posts -> qualified leads appear on live dashboard with relevance score and AI insight -> instantly admin notified via email.
- Real-time dashboard updates via WebSockets with match scoring and recommended action per lead
- Handled production challenges including async database architecture, Pydantic validation layers, Docker-based PostgreSQL setup, and CORS configuration
- Deployed on Cloudflare Pages (frontend), Hugging Face Spaces (backend), and Neon (PostgreSQL)

Live Demo: [Lead-Scout-Agent](https://lead-scout-agent.com)

AI HR Screening Agent

Next.js, FastAPI, PostgreSQL, LangGraph, OpenAI, PyPDF, LangChain

- Built as a full-stack AI system that automates the candidate screening process end-to-end
HR uploads candidate CV -> AI generates role-specific interview questions -> conducts structured text-based interview -> evaluation summary with rating and recommendation generated -> HR reviews final report
- Built interview flow as a LangGraph multi-phase state machine greeting -> intro -> interview -> analysis
- Voice integration (STT/TTS) architecture designed and ready, pending API access

Live Demo: [HR Agent](https://hr-agent.com)

TECHNICAL SKILLS

- **Languages:** Python (primary), TypeScript
 - **AI Frameworks:** LangChain, LangGraph, MCP, RAG, CRAG, AI Agents, Tool Calling, State Graphs, Prompt Engineering, Structured Output using pydantic schemas, Guardrails
 - **LLM Providers:** OpenAI, Groq, Gemini, OpenRouter
 - **Memory:** In-context memory, Vector store retrieval (pgvector, Pinecone, FAISS), Persistent state (PostgreSQL, MySQL)
 - **Backend & APIs:** FastAPI, REST APIs, WebSockets, SQLAlchemy, Pydantic Schemas
 - **Frontend:** Next.js, React
 - **Databases:** PostgreSQL, MySQL (familiar), SQLite (familiar)
 - **Vector Databases:** Pinecone, FAISS, pgvector
 - **Deployment & Cloud:** AWS, Vercel, FastAPI Cloud, Cloudflare Pages, Hugging Face Spaces, Neon, Docker
 - **Tools:** Git, GitHub
 - **Additional Knowledge:** Machine Learning, Deep Learning, Pandas, NumPy, Data Visualization (Matplotlib, Seaborn, Plotly)
-

EDUCATION

Bachelor's Degree, Artificial Intelligence

University of Haripur (UOH)

2021–2025

CERTIFICATIONS & COURSES

CampusX — Data Science Mentorship Program (DSMP 1.0)

CampusX — Data Science Mentorship Program (DSMP 2.0)

Completed 2025

In Progress